

CertFYI - Sample Presentation Content

Architecture Overview

CertFYI consists of a Next.js frontend, NestJS backend, PostgreSQL database, IPFS (Pinata), and an Ethereum Sepolia smart contract. The blockchain is the source of truth while PostgreSQL acts as a searchable off-chain index.

Authentication (SIWE)

Users authenticate using Sign-In with Ethereum (SIWE). Wallet ownership is verified by signing a message without requiring a blockchain transaction or gas fees.

Smart Contract

Stores the document hash, enforces role-based access control (Admin/Issuer), and records the immutable block timestamp.

Challenge

Ensuring Secure Role-Based Authorization for Certificate Management.

Future Improvements

- Subscription Model – Flexible plans for institutions.
- Bulk Certificate Issuance – Upload CSV to issue thousands of certificates.
- Cross-Chain Certificate Verification – Verify certificates seamlessly across multiple blockchain networks.
- AI Fraud Detection – Detect duplicate or suspicious certificates automatically.